FINAL YEAR PROJECT PROPOSAL

Smart CV Analyzer and Automated Candidate Shortlisting System

Analyze CVs. Match job requirements. Simplify shortlisting.

Submitted by

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1. Introduction

1.1 Background

Companies often receive many CVs when they announce a job vacancy. Each CV may contain different information about education, work experience, skills and qualifications. Reviewing every CV manually can take a great deal of time, especially when the number of applicants is high. It can also be difficult for recruiters to compare candidates consistently.

1.2 Project Overview

The Smart CV Analyzer and Automated Candidate Shortlisting System is a web-based application that will help recruiters perform the initial screening of job applications. A recruiter will enter the job requirements, such as minimum education, required experience and relevant skills, and upload multiple CVs. The application will extract information from the CVs, compare it with the requirements, categorize the results and provide short candidate summaries for easier review.

1.3 Importance of the Project

The project applies text processing and web application development to a common recruitment task. It is intended to reduce repetitive manual work and present candidate information in a clear and organized format. The system will support recruiters; it will not replace human judgment or make final hiring decisions.

2. Problem Statement

When a company receives a large number of applications for one vacancy, HR staff must read and compare each CV with the job description. This process can be slow and repetitive. Important information may be overlooked, and comparing candidates in different CV formats can be difficult. Companies need a simple tool that can identify relevant details, compare them with job requirements and organize applications so that recruiters can review potentially suitable candidates more efficiently.

3. Proposed Solution

The proposed system will provide a single interface where a recruiter can enter job requirements and upload multiple CVs in PDF format. The application will extract readable text and identify relevant details, including education, work experience and skills. It will compare the extracted information with the requirements and place each CV into a matching, partially matching or needs-review category. A concise summary will show the candidate's key qualifications, matching skills and any requirements that appear to be missing or unclear.

The system will explain why a CV was placed in a category instead of presenting only a score. Missing information will be marked as unknown or requiring review rather than automatically treated as proof that a candidate is unqualified. Recruiters will remain responsible for reviewing results and making shortlisting decisions.

4. Project Objectives

4.1 General Objective

To develop a web-based application that analyzes multiple CVs against company-defined job requirements and organizes the results to make initial candidate screening easier and faster.

4.2 Specific Objectives

1. Allow recruiters to enter job requirements for education, experience and skills.

2. Allow multiple CVs to be uploaded in PDF format.

3. Extract text and relevant candidate details from uploaded CVs.

4. Compare candidate details with the specified job requirements.

5. Categorize CVs into matching, partially matching and needs-review groups.

6. Generate short, readable summaries of candidate qualifications.

7. Show matching criteria and missing or unclear information.

8. Keep the system easy to use and protect uploaded CVs from unnecessary access.

5. Scope of the Project

5.1 In Scope

• Recruiter interface for entering job title and requirements.

• Upload and processing of multiple text-based PDF CVs.

• Extraction of skills, education and experience where the CV text allows it.

• Requirement matching using rules and text-processing techniques.

• Categorization and sorting of CVs according to their match with the job.

• Short candidate summaries and explanations of matching or missing criteria.

• Basic testing using sample CVs and sample job descriptions.

5.2 Out of Scope

• Making final hiring or rejection decisions without human review.

• Guaranteeing that every CV format or scanned image will be read perfectly.

• Connecting directly to commercial job portals or company HR systems in the first version.

• Using sensitive or unrelated personal characteristics to rank candidates.

• Providing a legally validated assessment of a candidate's suitability.

5.3 Future Enhancements

Future versions may support scanned CVs through OCR, additional file formats, recruiter accounts, downloadable shortlisting reports, more advanced semantic matching and integration with existing recruitment systems.

6. Target Users

The primary users will be HR officers, recruiters and small or medium-sized companies that need to screen multiple applications. Recruitment agencies and university career offices may also benefit from a tool that organizes CVs against a particular job or internship description. The application is intended to assist users who have basic computer skills.

7. Proposed Features

7.1 Job Requirement Input

The recruiter will enter the job title and define the required education, minimum experience and skills. Requirements should be written clearly so the system can compare them with candidate information.

7.2 Multiple CV Upload

The user will upload multiple PDF CVs in one session. The application will check file type and handle files that cannot be read by showing an appropriate message.

7.3 CV Text and Information Extraction

The application will extract text from supported PDFs and identify details such as education, work experience and skills. The quality of extraction may vary depending on the layout and readability of each CV.

7.4 Requirement Matching and Filtering

The system will compare candidate details with the job requirements. Results will be organized into matching, partially matching and needs-review categories. The recruiter will be able to see which criteria matched and which information was missing or unclear.

7.5 Candidate Summary

For each CV, the system will display a short summary of relevant education, experience, skills and matching details. This will help recruiters review the most relevant information without repeatedly searching through the full document.

7.6 Review and Export

The first version may provide sorting and filtering of results and, if time permits, an option to export the shortlist or summaries. All results will be available for human review before any recruitment decision is made.

8. Proposed Technologies

Technology	Purpose
Python	Main programming language for application logic and processing.
Streamlit	Creating a simple web interface for recruiters.
PyMuPDF	Extracting text from supported PDF CVs.
NLP / text-processing techniques	Identifying skills and comparing CV content with job requirements.
Pandas (optional)	Organizing candidate details and screening results.
SQLite (optional)	Storing job requirements and results if persistent storage is needed.
Git and GitHub	Version control and hosting the project source code.

The final technology choices may be adjusted after implementation begins and the requirements are reviewed with the project supervisor.

9. Development Methodology

The project will follow an iterative development approach. First, the requirements and sample job descriptions will be defined. Next, the interface and CV upload process will be developed. Text extraction and requirement matching will then be implemented and tested using sample CVs. After that, the results will be presented in categorized lists and summaries. The application will be improved through testing and feedback, and the final stage will include documentation and a demonstration.

10. Expected Outcomes

The expected outcome is a working prototype that accepts job requirements and multiple supported PDF CVs, extracts relevant information, compares candidates with the requirements, categorizes the results and presents readable summaries. The application should reduce repetitive manual screening work and help recruiters locate relevant information more quickly. Its usefulness will be assessed through tests using sample CVs and job descriptions.

11. Project Significance

The project is significant because it addresses a practical recruitment problem using accessible software technologies. It gives the student experience in Python programming, document processing, text analysis, web interface development, testing and version control. For recruiters, it offers a more organized first-level screening process and clearer reasons for reviewing particular CVs.

12. Project Feasibility

Technical Feasibility: Python, Streamlit and PDF text-extraction libraries are accessible and suitable for a student prototype. The initial system can use rule-based and text-based matching without requiring expensive hardware.

Operational Feasibility: The interface will focus on a straightforward workflow: enter requirements, upload CVs, view categorized results and review summaries.

Economic Feasibility: The proposed tools are open-source or have free options, so development can be completed with limited software cost.

Schedule Feasibility: The project can be developed in stages, beginning with PDF extraction and basic matching, then adding summaries and improvements as time allows.

13. Tentative Project Timeline

Phase	Activity	Expected Deliverable
1	Finalize requirements and review sample CVs	Requirements list
2	Set up project and user interface	Basic Streamlit application
3	Implement PDF upload and text extraction	CV text extraction module
4	Implement requirement input and matching rules	Matching and filtering module
5	Generate summaries and result categories	Candidate results screen
6	Test with sample CVs, fix issues and document	Tested prototype and report

The timeline is tentative and can be adjusted according to the university's project schedule and supervisor feedback.

14. Ethical and Privacy Considerations

CVs contain personal information. Testing should use sample or consented CVs, and uploaded files should not be shared unnecessarily. The system should avoid using sensitive personal characteristics such as religion, gender or ethnicity for ranking candidates. It should display matching reasons clearly, acknowledge missing information and leave final decisions to human reviewers. Results may contain extraction errors, so recruiters should verify important details in the original CV.

15. Conclusion

The Smart CV Analyzer and Automated Candidate Shortlisting System is proposed as a practical application for organizing the initial screening of job applications. By allowing recruiters to define job requirements, upload multiple CVs, compare qualifications and view concise candidate summaries, the system aims to reduce manual effort and make CV review more efficient. The application will assist rather than replace HR judgment, and the final decisions will remain with the recruiter.

16. References

• Python Documentation: https://docs.python.org/3/

• Streamlit Documentation: https://docs.streamlit.io/

• PyMuPDF Documentation: https://pymupdf.readthedocs.io/

• spaCy Documentation: https://spacy.io/ (if spaCy is used)

• GitHub Documentation: https://docs.github.com/

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